

Lecture Six

1 Hyperbolic Functions

1.1 Hyperbolic Sine Function

We define the hyperbolic function as follows:

$$y = \sinh x = \frac{e^x - e^{-x}}{2}$$

The domain of definition is $D = \mathbb{R}$. To determine the range, let $u = e^x$

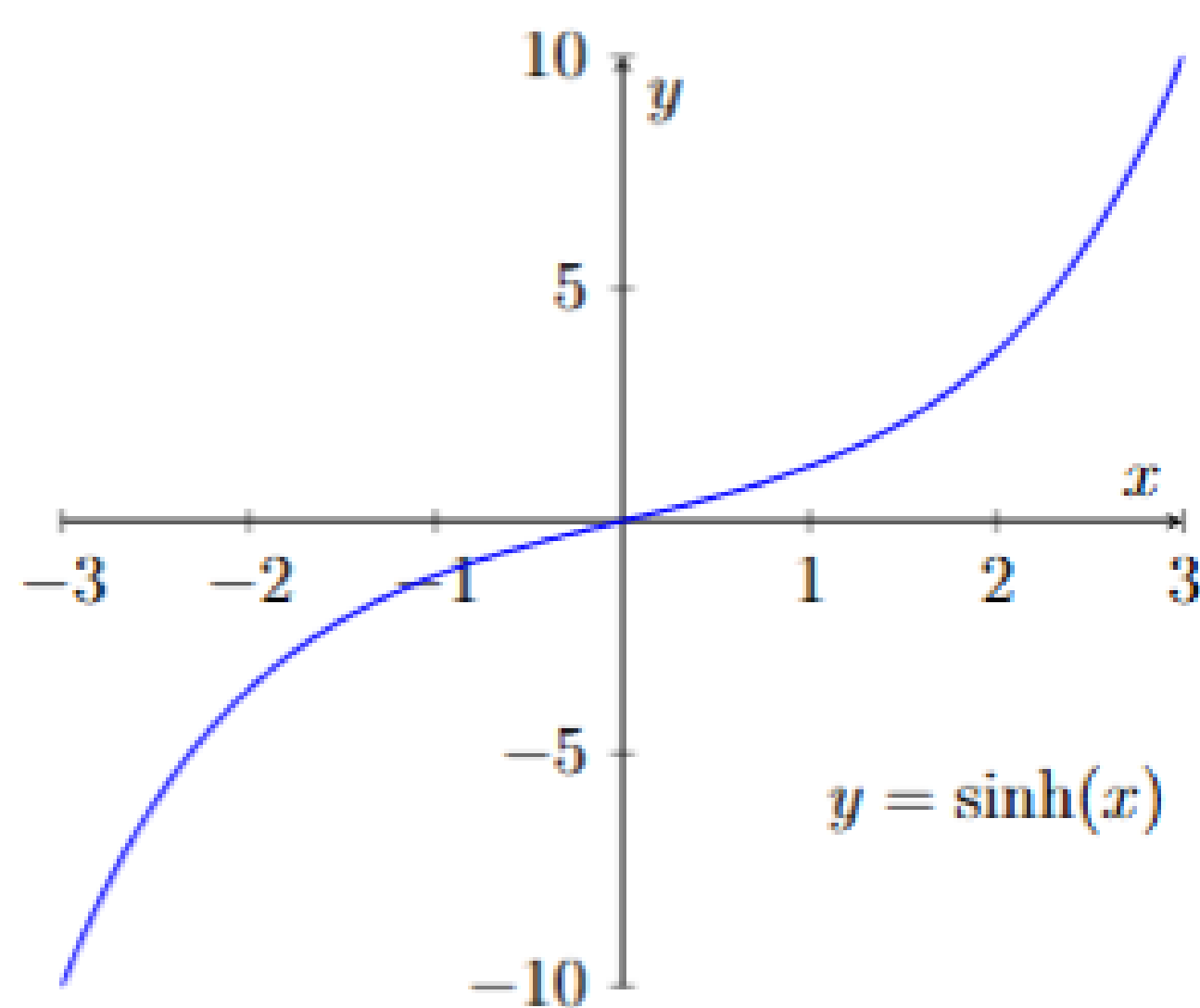
$$\begin{aligned} \Rightarrow y &= \frac{u - \frac{1}{u}}{2}, u > 0 \Rightarrow u^2 - 2yu - 1 = 0 \\ \Rightarrow u &= \frac{2y \pm \sqrt{4y^2 + 4}}{2} = y \pm \sqrt{y^2 + 1} \end{aligned}$$

Since $\sqrt{y^2 + 1} > y$ and $u > 0$ this implies that $\boxed{u = y + \sqrt{y^2 + 1}}$. Clearly $u > 0, \forall y \in \mathbb{R} \Rightarrow \mathbf{R} = \mathbb{R}$

1.1.1 Properties

1. The hyperbolic sine function is odd since

$$\begin{aligned} \sinh(-x) &= \frac{e^{-x} - e^x}{2} \\ &= -\left(\frac{e^x - e^{-x}}{2}\right) = -\sinh(x) \end{aligned}$$



$$2. \sinh(0) = \frac{e^0 - e^0}{2} = \frac{1 - 1}{2} = 0.$$

$$3. \lim_{x \rightarrow -\infty} \sinh x = \lim_{x \rightarrow -\infty} \left[\frac{e^x - e^{-x}}{2} \right] \rightarrow \frac{0 - \infty}{2} \rightarrow -\infty.$$

$$4. \lim_{x \rightarrow \infty} \sinh(x) = \lim_{x \rightarrow \infty} \left[\frac{e^x - e^{-x}}{2} \right] \rightarrow \frac{\infty - 0}{2} \rightarrow \infty$$

5. If x is **large** then $e^{-x} \rightarrow 0$ and this implies that $\sinh x \rightarrow \frac{e^x}{2}$.

6. If x is very small, then $e^x \rightarrow 0$ and $\sinh x \rightarrow -\frac{e^{-x}}{2}$.

1.2 Hyperbolic Cosine Function

We define hyperbolic cosine function as follows:

$$y = \cosh x = \frac{e^x + e^{-x}}{2}$$

and clearly $D = \mathbb{R}$. Now to determine the range, let $u = e^x$

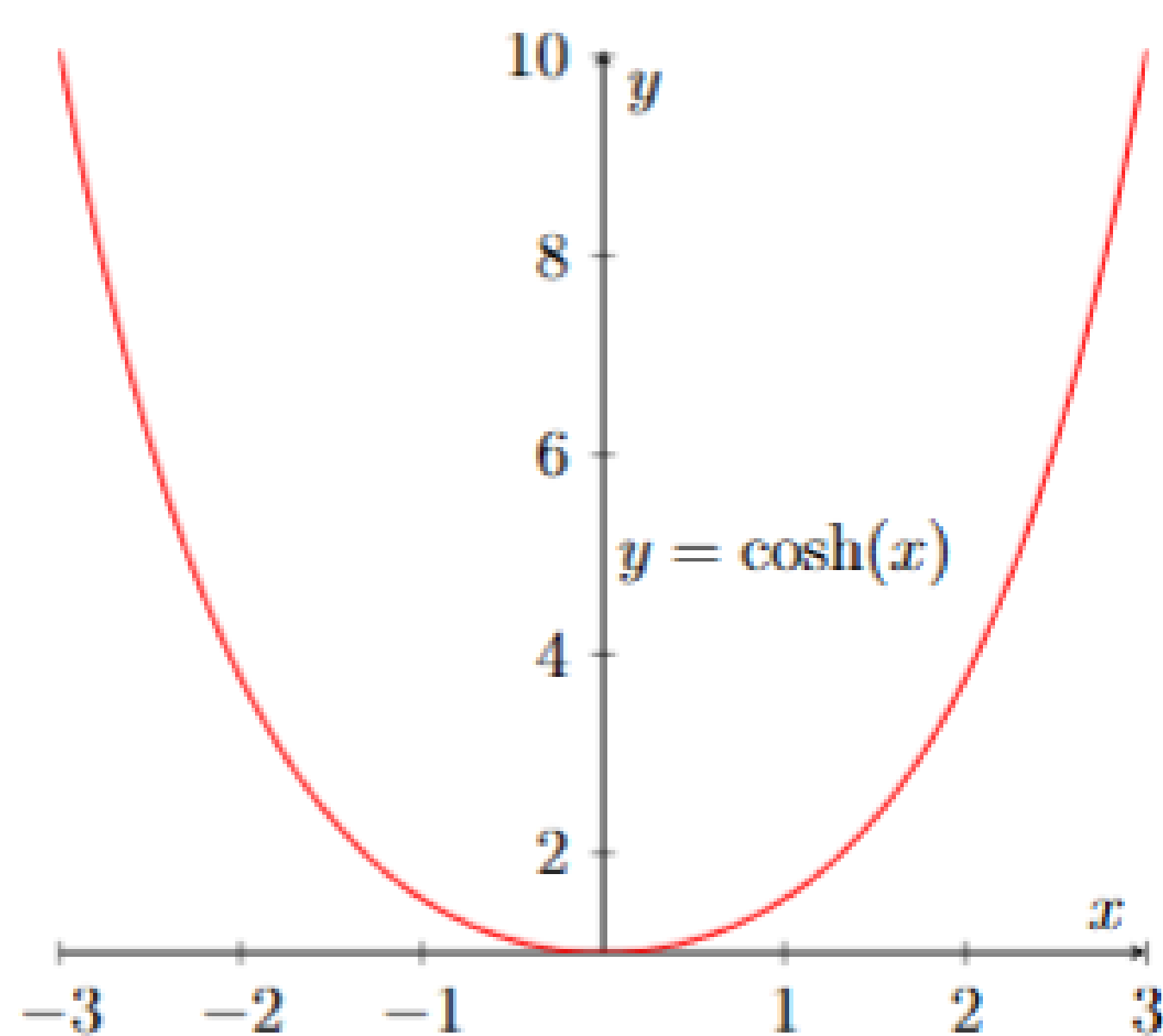
$$\begin{aligned} \Rightarrow y &= \frac{u + \frac{1}{u}}{2}, u > 0 \Rightarrow u^2 - 2yu + 1 = 0 \\ \Rightarrow u &= \frac{2y \pm \sqrt{4y^2 - 4}}{2} = y \pm \sqrt{y^2 - 1} \end{aligned}$$

$u > 0$ if $y > 0$ and $|y| \geq 1$, therefore the range will be $\mathbf{R} = [1, \infty)$

1.2.1 Properties

1. The hyperbolic cosine function is even since

$$\begin{aligned} \cosh(-x) &= \frac{e^{-x} + e^x}{2} \\ &= \left(\frac{e^x + e^{-x}}{2} \right) = \cosh(x) \end{aligned}$$



$$2. \cosh(0) = \frac{e^0 + e^0}{2} = \frac{1+1}{2} = 1.$$

$$3. \lim_{x \rightarrow -\infty} \cosh x = \lim_{x \rightarrow -\infty} \left[\frac{e^x + e^{-x}}{2} \right] \rightarrow \frac{0 + \infty}{2} \rightarrow \infty.$$

$$4. \lim_{x \rightarrow \infty} \cosh(x) = \lim_{x \rightarrow \infty} \left[\frac{e^x + e^{-x}}{2} \right] \rightarrow \frac{\infty + 0}{2} \rightarrow \infty$$

5. If x is **large** then $e^{-x} \rightarrow 0$ and this implies that $\cosh x \rightarrow \frac{e^x}{2}$.

6. If x is very small, then $e^x \rightarrow 0$ and $\cosh x \rightarrow \frac{e^{-x}}{2}$.

1.3 Hyperbolic Tangent function

$$y = \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$

$D = \mathbb{R}$. Now for the range, let $u = e^x$, we get:

$$y = \frac{u^2 - 1}{u^2 + 1}, u > 0 \Rightarrow u^2 = \frac{y+1}{1-y} \Rightarrow u = \sqrt{\frac{y+1}{1-y}}$$

This implies that $u > 0$ if $(y+1 > 0) \wedge (1-y > 0)$

$$\Rightarrow (y > -1) \cap (y < 1)$$

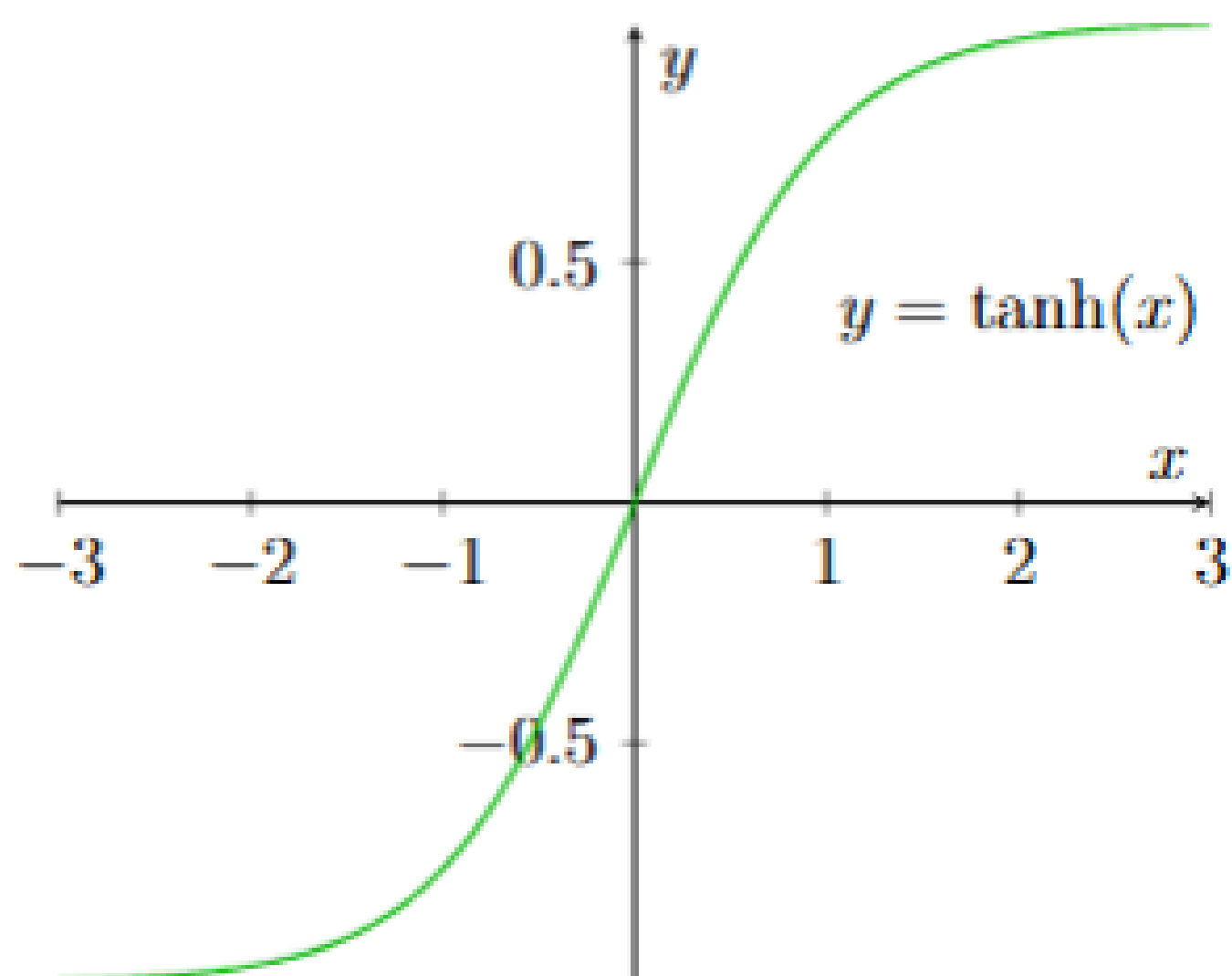
$$\Rightarrow (-1 < y < 1) \Rightarrow y \in (-1, 1)$$

Or $(y+1 < 0) \wedge (1-y < 0) \Rightarrow (y < -1) \cap (y > 1) = \emptyset$. Therefore $\mathbf{R} = (-1, 1)$.

1.3.1 Properties

1. The hyperbolic tangent function is odd since

$$\begin{aligned}\tanh(-x) &= \frac{\sinh(-x)}{\cosh(-x)} \\ &= \frac{-\sinh(x)}{\cosh(x)} = -\tanh(x)\end{aligned}$$



$$2. \tanh(0) = \frac{e^0 - e^0}{e^0 + e^0} = \frac{1 - 1}{1 + 1} = \frac{0}{2} = 0.$$

$$3. \lim_{x \rightarrow -\infty} \tanh x = \lim_{x \rightarrow -\infty} \left[\frac{e^{2x} - 1}{e^{2x} + 1} \right] \rightarrow \frac{0 - 1}{0 + 1} \rightarrow -1.$$

$$4. \lim_{x \rightarrow \infty} \tanh(x) = \lim_{x \rightarrow \infty} \left[\frac{1 - e^{-2x}}{1 + e^{-2x}} \right] \rightarrow \frac{1 - 0}{1 + 0} \rightarrow 1$$

$$5. \text{ If } x \text{ is large then } \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \rightarrow \frac{e^x - 0}{e^x + 0} \rightarrow \frac{e^x}{e^x} \rightarrow 1.$$

$$6. \text{ If } x \text{ is very small, then } \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \rightarrow \frac{0 - e^{-x}}{0 + e^{-x}} \rightarrow \frac{-e^{-x}}{e^{-x}} \rightarrow -1.$$

1.4 Hyperbolic Cotangent Function

$$y = \coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{e^{2x} + 1}{e^{2x} - 1} = \frac{1 + e^{-2x}}{1 - e^{-2x}}$$

$D = \mathbb{R} - \{0\}$. Now for the range, let $u = e^x$, we get:

$$y = \frac{u^2 + 1}{u^2 - 1}, u > 0 \Rightarrow u^2 = \frac{y + 1}{y - 1} \Rightarrow u = \sqrt{\frac{y + 1}{y - 1}}$$

This implies that $u > 0$ if $(y + 1 > 0) \wedge (y - 1 > 0)$

$$\begin{aligned}\Rightarrow (y > -1) \cap (y > 1) &= y > 1 \\ \Rightarrow y &\in (1, \infty)\end{aligned}$$

Or $(y + 1 < 0) \wedge (y - 1 < 0)$

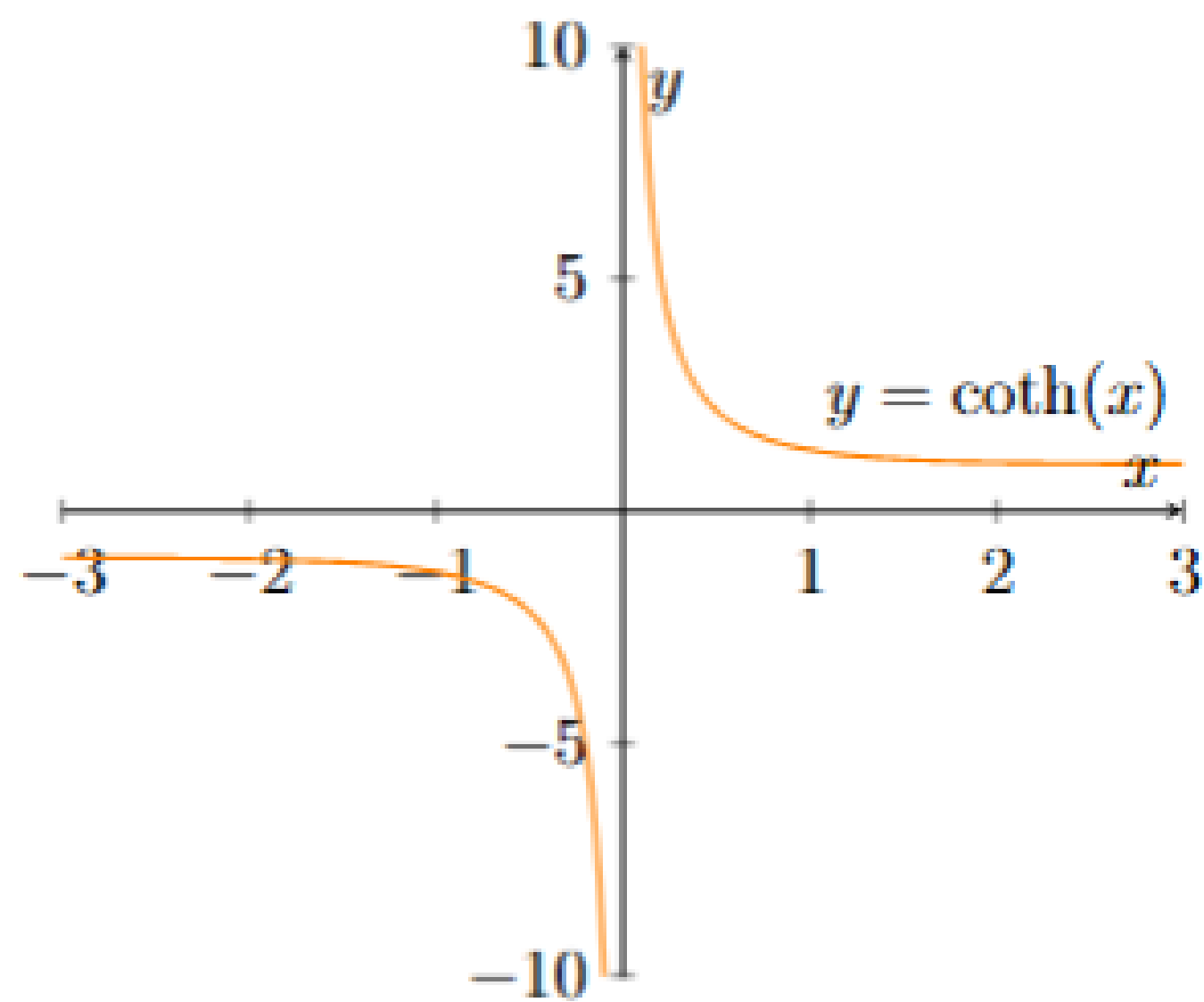
$$\begin{aligned}\Rightarrow (y < -1) \cap (y < 1) &= y < -1 \\ \Rightarrow y &\in (-\infty, -1)\end{aligned}$$

Therefore the range is $\mathbf{R} = (-\infty, -1) \cup (1, \infty) \equiv \mathbb{R} - [-1, 1]$

1.4.1 Properties

1. The hyperbolic cotangent function is odd since

$$\begin{aligned}\coth(-x) &= \frac{\cosh(-x)}{\sinh(-x)} \\ &= \frac{\cosh(x)}{-\sinh(x)} = -\coth(x)\end{aligned}$$



2. $\lim_{x \rightarrow 0^-} \coth(x) = \lim_{h \rightarrow 0} \frac{1}{\tanh(-h)} \rightarrow -\lim_{h \rightarrow 0} \frac{1}{\tanh(h)} \rightarrow \frac{-1}{0} \rightarrow -\infty.$
3. $\lim_{x \rightarrow 0^+} \coth(x) = \lim_{h \rightarrow 0} \frac{1}{\tanh(h)} \rightarrow \lim_{h \rightarrow 0} \frac{1}{\tanh(h)} \rightarrow \frac{1}{0} \rightarrow \infty.$
4. $\lim_{x \rightarrow -\infty} \coth(x) = \lim_{x \rightarrow \infty} \left[\frac{e^{2x} + 1}{e^{-2x} - 1} \right] \rightarrow \frac{0 + 1}{0 - 1} \rightarrow -1$
5. $\lim_{x \rightarrow \infty} \coth(x) = \lim_{x \rightarrow \infty} \left[\frac{1 + e^{-2x}}{1 - e^{-2x}} \right] \rightarrow \frac{1 + 0}{1 - 0} \rightarrow 1$

1.5 Hyperbolic Secant Function

$$y = \operatorname{sech}(x) = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

Clearly $D = \mathbb{R}$. To determine the range, let $u = e^x$

$$\Rightarrow y = \frac{2}{u + \frac{1}{u}} = \frac{2u}{u^2 + 1} \Rightarrow yu^2 - 2u + y = 0 \quad ; y \neq 0$$

$$\Rightarrow u = \frac{2 \pm \sqrt{4 - 4y^2}}{2y} = \frac{1 \pm \sqrt{1 - y^2}}{y}$$

Since $u > 0$, either $(y > 0) \wedge (1 \pm \sqrt{1 - y^2} > 0)$

$$\Rightarrow (y > 0) \wedge (y^2 \leq 1) \Rightarrow y \in (0, \infty) \cap y \in [-1, 1]$$

$$\Rightarrow y \in (0, 1]$$

Or $(y < 0) \wedge (1 \pm \sqrt{1 - y^2} < 0)$ and its impossible, therefore $\mathbf{R} = (0, 1]$

1.5.1 Properties

1. The hyperbolic secant function is even since

$$\operatorname{sech}(-x) = \frac{1}{\cosh(-x)} = \frac{1}{\cosh(x)} = \operatorname{sech}(x)$$

2. $\lim_{x \rightarrow 0} \operatorname{sech}(x) = \lim_{x \rightarrow 0} \frac{1}{\cosh x} = \frac{1}{1} = 1$
3. $\lim_{x \rightarrow -\infty} \operatorname{sech}(x) = \lim_{x \rightarrow -\infty} \frac{1}{\cosh x} \rightarrow \frac{2}{\infty} \rightarrow 0$
4. $\lim_{x \rightarrow \infty} \operatorname{sech}(x) = \lim_{x \rightarrow \infty} \frac{1}{\cosh x} \rightarrow \frac{2}{\infty} \rightarrow 0$

1.6 Hyperbolic Cosecant Function

$$y = \operatorname{csch}(x) = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

Clearly $D = \mathbb{R} - \{0\}$. To determine the range, let $u = e^x$

$$\begin{aligned} \Rightarrow y &= \frac{2}{u - \frac{1}{u}} = \frac{2u}{u^2 - 1} \Rightarrow yu^2 - 2u - y = 0 \quad ; y \neq 0 \\ \Rightarrow u &= \frac{2 \pm \sqrt{4 + 4y^2}}{2y} = \frac{1 \pm \sqrt{1 + y^2}}{y} \end{aligned}$$

1. Either $(y > 0) \wedge (1 \pm \sqrt{1 + y^2} > 0)$, this implies that:

$$(y > 0) \wedge (1 + \sqrt{1 + y^2} > 0) \Rightarrow y \in \mathbb{R}^+$$

$$\text{or } (y > 0) \wedge (1 - \sqrt{1 + y^2} > 0)$$

$$\Rightarrow (y > 0) \wedge (\sqrt{1 + y^2} < 1)$$

and its impossible, thus $y \in \mathbb{R}^+$

2. Or $(y < 0) \wedge (1 \pm \sqrt{1 + y^2} < 0)$, this implies that

$$(y < 0) \wedge (1 + \sqrt{1 + y^2} < 0)$$

and its impossible. or

$$(y < 0) \wedge (1 - \sqrt{1 + y^2} < 0) \Rightarrow y \in \mathbb{R}^-$$

Therefore

$$\mathbf{R} = \mathbb{R}^+ \cup \mathbb{R}^- = \mathbb{R} - \{0\}$$

1.6.1 Properties

1. The hyperbolic cosecant function is odd since

$$\operatorname{csch}(-x) = \frac{1}{\sinh(-x)} = \frac{1}{-\sinh(x)} = -\operatorname{csch}(x)$$

.

2. $\lim_{x \rightarrow 0^-} \operatorname{csch}(x) = \lim_{h \rightarrow 0} \frac{1}{\sinh(-h)} \rightarrow -\lim_{h \rightarrow 0} \frac{1}{\sinh(h)} \rightarrow \frac{-1}{0} = -\infty.$

$$3. \lim_{x \rightarrow 0^+} \operatorname{csch}(x) = \lim_{h \rightarrow 0} \frac{1}{\sinh(h)} \rightarrow \frac{1}{0} \rightarrow \infty.$$

$$4. \lim_{x \rightarrow -\infty} \operatorname{csch}(x) = \lim_{x \rightarrow \infty} \left[\frac{2}{e^x - e^{-x}} \right] \rightarrow \frac{2}{0 - \infty} \rightarrow 0$$

$$5. \lim_{x \rightarrow \infty} \operatorname{csch}(x) = \lim_{x \rightarrow \infty} \left[\frac{2}{e^x - e^{-x}} \right] \rightarrow \frac{2}{\infty - 0} \rightarrow 0$$

1.7 Hyperbolic Functions Identities

$$1. \forall x \in \mathbb{R}, \cosh^2 x - \sinh^2 x = 1$$

Proof

$$\left[\frac{e^x + e^{-x}}{2} \right]^2 - \left[\frac{e^x - e^{-x}}{2} \right]^2 = \frac{e^{2x} + 2 + e^{-2x} - (e^{2x} + 2 - e^{-2x})}{4} = \frac{4}{4} = 1$$

2.

$$\forall x \in \mathbb{R}, \sinh^2 x = \cosh^2 x - 1$$

$$\forall x \in \mathbb{R}, 1 - \tanh^2 x = \operatorname{sech}^2(x)$$

$$\forall x \in \mathbb{R}, (x \neq 0) \quad \coth^2(x) - 1 = \operatorname{csch}^2(x)$$

$$\forall x \in \mathbb{R}, \cosh x + \sinh x = e^x.$$

$$3. \quad (i) \sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

Proof

The R.H.S

$$\begin{aligned} &\Rightarrow \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^y + e^{-y}}{2} \right) + \left(\frac{e^x + e^{-x}}{2} \right) \left(\frac{e^y - e^{-y}}{2} \right) \\ &\Rightarrow \frac{e^{x+y} + e^{x-y} - e^{y-x} - e^{-(x+y)}}{4} + \frac{e^{x+y} - e^{x-y} + e^{y-x} - e^{-(x+y)}}{4} \\ &= \frac{2[e^{x+y} - e^{-(x+y)}]}{4} = \frac{e^{x+y} - e^{-(x+y)}}{2} = \sinh(x + y) \quad \square \end{aligned}$$

$$(ii) \sinh(x - y) = \sinh x \cosh y - \cosh x \sinh y$$

$$(iii) \cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$$

$$(iv) \tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \mp \tanh x \tanh y}$$

$$4. \quad (i) \sinh 2x = 2 \sinh(x) \cosh(x)$$

$$(ii) \cosh 2x = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 2 \sinh^2 x + 1$$

1.8 Derivative of Hyperbolic Function

$$1. \frac{d}{dx}[\sinh x] = \cosh x$$

Proof

$$\frac{d}{dx}[\sinh x] = \frac{d}{dx} \left[\frac{e^x - e^{-x}}{2} \right] = \frac{1}{2} \frac{d}{dx} [e^x - e^{-x}] = \frac{1}{2} [e^x + e^{-x}] = \left[\frac{e^x + e^{-x}}{2} \right] = \cosh x$$

$$2. \frac{d}{dx}[\cosh x] = \sinh x$$

Proof

$$\frac{d}{dx}[\cosh x] = \frac{d}{dx} \left[\frac{e^x + e^{-x}}{2} \right] = \frac{1}{2} \frac{d}{dx} [e^x + e^{-x}] = \frac{1}{2} [e^x - e^{-x}] = \left[\frac{e^x - e^{-x}}{2} \right] = \sinh x$$

$$3. \frac{d}{dx}[\tanh(x)] = \operatorname{sech}^2(x)$$

Proof

$$\frac{d}{dx}[\tanh(x)] = \frac{d}{dx} \left[\frac{\sinh x}{\cosh x} \right] = \frac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x$$

$$4. \frac{d}{dx}[\coth(x)] = \operatorname{csch}^2(x)$$

Proof

$$\frac{d}{dx}[\coth(x)] = \frac{d}{dx} \left[\frac{\cosh x}{\sinh x} \right] = \frac{\sinh x \sinh x - \cosh x \cosh x}{\sinh^2 x} = \frac{-1}{\sinh^2 x} = -\operatorname{csch}^2 x$$

$$5. \frac{d}{dx}[\operatorname{sech}(x)] = -\operatorname{sech}(x) \tanh(x)$$

Proof

$$\frac{d}{dx}[\operatorname{sech}(x)] = \frac{d}{dx} [(\cosh x)^{-1}] = -(\cosh x)^{-2} \times \sinh x = -\operatorname{sech}(x) \tanh(x)$$

$$6. \frac{d}{dx}[\operatorname{csch}(x)] = -\operatorname{csch}(x) \coth(x)$$

Proof

$$\frac{d}{dx}[\operatorname{csch}(x)] = \frac{d}{dx} [(\sinh x)^{-1}] = -(\sinh x)^{-2} \times \cosh x = -\operatorname{csch}(x) \coth(x)$$

In general

No.	The Hyp. Function	Derivative
(1)	$\frac{d}{dx}[\sinh(u(x))]$	$u'(x) \cosh(u(x))$
(2)	$\frac{d}{dx}[\cosh(u(x))]$	$u'(x) \sinh(u(x))$
(3)	$\frac{d}{dx}[\tanh(u(x))]$	$u'(x) \operatorname{sech}^2(u(x))$
(4)	$\frac{d}{dx}[\coth(u(x))]$	$-u'(x) \operatorname{csch}^2(u(x))$
(5)	$\frac{d}{dx}[\operatorname{sech}(u(x))]$	$-u'(x) \operatorname{sech}(u(x)) \tanh(u(x))$
(6)	$\frac{d}{dx}[\operatorname{csch}(u(x))]$	$-u'(x) \operatorname{csch}(u(x)) \coth(u(x))$

Example 1.8.1 Evaluate the following limits

$$1. \lim_{x \rightarrow 0} \frac{\sinh(\alpha x)}{x}$$

$$2. \lim_{x \rightarrow 0} \frac{\cosh(\alpha x) - 1}{x^2}$$

$$3. \lim_{x \rightarrow 0} \frac{\cosh(\alpha x) - \cosh(\beta x)}{x^2}$$

Sol

Before we solve this limits, note that the hyperbolic functions can be expressed as an infinite series as follows:

We know

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^k}{k!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

and

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots + (-1)^k \frac{x^k}{k!} + \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{x^k}{k!}$$

Now

$$\begin{aligned} \cosh x &= \frac{1}{2}(e^x + e^{-x}) \\ &= \frac{1}{2} \left[\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^k}{k!} + \cdots \right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots + (-1)^k \frac{x^k}{k!} + \cdots \right) \right] \\ &= \frac{1}{2} \left[2 + 2\frac{x^2}{2!} + 2\frac{x^4}{4!} + \cdots + 2\frac{x^{2k}}{(2k)!} + \cdots \right] = \left[1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2k}}{(2k)!} + \cdots \right] \end{aligned}$$

$$\therefore \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2k}}{(2k)!} + \cdots = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$$

As same as $\cosh x$, we see

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2k+1}}{(2k+1)!} + \cdots = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

Now:

$$1. \lim_{x \rightarrow 0} \frac{\sinh(\alpha x)}{x}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\left(\alpha x + \frac{\alpha^3 x^3}{3!} + \frac{\alpha^5 x^5}{5!} + \dots \right)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\alpha x \left(1 + \frac{\alpha^2 x^2}{3!} + \frac{\alpha^4 x^4}{5!} + \dots \right)}{x} \\ &= \alpha \lim_{x \rightarrow 0} \left(1 + \frac{\alpha^2 x^2}{3!} + \frac{\alpha^4 x^4}{5!} + \dots \right) = \alpha(1) = \alpha. \end{aligned}$$

$$2. \lim_{x \rightarrow 0} \frac{\cosh(\alpha x) - 1}{x^2}$$

$$\cosh(\alpha x) - 1 = \left(1 + \frac{\alpha^2 x^2}{2!} + \frac{\alpha^4 x^4}{4!} + \dots \right) - 1 = \frac{\alpha^2 x^2}{2!} + \frac{\alpha^4 x^4}{4!} + \dots$$

Now:

$$\frac{\cosh(\alpha x) - 1}{x^2} = \frac{\alpha^2}{2!} + \frac{\alpha^2 x^2}{4!} + \dots$$

Therefore

$$\lim_{x \rightarrow 0} \frac{\cosh(\alpha x) - 1}{x^2} = \lim_{x \rightarrow 0} \left[\frac{\alpha^2}{2!} + \frac{\alpha^2 x^2}{4!} + \dots \right] = \frac{\alpha^2}{2!}.$$

3. Exercise.

Example 1.8.2 Find $\frac{dy}{dx}$ for the following functions.

$$1. y = \cosh(2x) \sinh(x)$$

Sol

$$\frac{dy}{dx} = \cosh(2x) \times \cosh(x) + \sinh(x) \times (2 \sinh(2x))$$

$$2. y = \operatorname{sech}(\operatorname{csch}(x))$$

Sol

$$\frac{dy}{dx} = -\operatorname{csch}(x) \coth(x) \times -\operatorname{sech}(\operatorname{csch}(x)) \tanh(\operatorname{csch}(x))$$

$$3. y = \tanh\left(\frac{2x}{1-x^2}\right)$$

Sol

$$\frac{dy}{dx} = \frac{2(1-x^2) + 4x}{(1-x^2)^2} \operatorname{sech}^2\left(\frac{2x}{1-x^2}\right)$$

$$4. y = \ln \sqrt{\frac{1+\tanh x}{1-\tanh x}} \text{ prove that } \frac{dy}{dx} = 1$$

Sol

Exercise

Exercise 1.8.1 1. Express the following functions as a hyperbolic functions

(i) $y = e^x + 4e^{-x}$

Sol

(ii) $y = e^{3x} + e^{-2x}$

Sol

2. Find $\frac{dy}{dx}$ for the following functions:

(i) $y = \sinh(\sqrt{x}) + \frac{\cosh x}{\sqrt{x}}$

Sol

(ii) $y = x \cosh(x + \tanh x)$

Sol

(iii) $y = x^2 \tanh(x + \sin x)$

Sol

$$(iv) \quad y = \frac{x^2 + \sin 2x}{x + 3 \sinh 4x}$$

Sol

$$(v) \quad y = \tan^{-1} \left[\frac{1 + \tanh x}{1 - \tanh x} \right]$$

Sol

$$(vi) \quad y + \tanh(xy) = e^{y-x}$$

Sol

3. Find the following limits:

$$(i) \quad \lim_{x \rightarrow 0} \frac{x - \sinh(\beta x)}{x}$$

Sol

$$(ii) \quad \lim_{x \rightarrow 0} \frac{e^{\alpha x} - 1}{\sinh(\beta x)}$$

Sol

$$(iii) \quad \lim_{x \rightarrow 0} \frac{\tanh(x)}{x}$$

Sol

$$(iv) \quad \lim_{x \rightarrow 0} \frac{e^{-2x} - 1}{\sinh(5x)}$$

Sol

2 Inverse Hyperbolic Functions

2.1 Inverse Hyperbolic Sine Function

We define it as:

$$y = \sinh^{-1} x \iff x = \sinh y = \frac{e^y - e^{-y}}{2}$$

Clearly $R = \mathbb{R}$. To determine the domain of definition, let $u = e^y \Rightarrow x = \frac{u - \frac{1}{u}}{2}$, $u > 0$. Now

we can write $u^2 - 2xu - 1 = 0 \Rightarrow u = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}$. Since $\sqrt{x^2 + 1} > x$ and $u > 0$, we get $u = x + \sqrt{x^2 + 1} \Rightarrow e^y = x + \sqrt{x^2 + 1}$

$$\therefore y = \ln \left(x + \sqrt{x^2 + 1} \right), \forall x \in \mathbb{R}$$

2.1.1 Derivative of Inverse Hyp. Sin

$y = \sinh^{-1} x = \ln \left(x + \sqrt{x^2 + 1} \right)$, then the first derivative is:

$$\frac{dy}{dx} = \left(1 + \frac{x}{\sqrt{x^2 + 1}} \right) \times \frac{1}{x + \sqrt{x^2 + 1}} = \left[\frac{x + \sqrt{x^2 + 1}}{\sqrt{x^2 + 1}} \right] \frac{1}{x + \sqrt{x^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}}$$

Therefore

$$\boxed{\frac{d}{dx}[\sinh^{-1} x] = \frac{1}{\sqrt{x^2 + 1}}}$$

Also we can differentiate it as:

$$y = \sinh^{-1} x \Rightarrow x = \sinh y \Rightarrow \frac{dx}{dy} = \cosh y \Rightarrow \frac{dy}{dx} = \frac{1}{\cosh y} = \frac{1}{\sqrt{\sinh^2 x + 1}} = \frac{1}{\sqrt{x^2 + 1}}$$

2.2 Inverse Hyperbolic Cosine Function

$y = \cosh^{-1} x \iff x = \cosh y = \frac{e^y + e^{-y}}{2}$. Clearly $R = [0, \infty)$, to determine the domain of definition let $u = e^y \Rightarrow x = \frac{u + \frac{1}{u}}{2}$, $u > 0 \Rightarrow u^2 - 2xu + 1 = 0$. Therefore

$$u = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}$$

$u > 0$ if $x \geq 1$. Since $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 - 1}) = 0$, we will take the value $(x + \sqrt{x^2 - 1})$ therefore $D = [1, \infty)$. Now:

$$u = x + \sqrt{x^2 - 1} \Rightarrow e^y = x + \sqrt{x^2 - 1} \Rightarrow y = \ln \left(x + \sqrt{x^2 - 1} \right), \forall x \in [0, \infty)$$

2.2.1 Derivative of Inverse Hyp. Cos

$y = \cosh^{-1} x = \ln \left(x + \sqrt{x^2 - 1} \right)$, then the first derivative is:

$$\frac{dy}{dx} = \left(1 + \frac{x}{\sqrt{x^2 - 1}} \right) \times \frac{1}{x + \sqrt{x^2 - 1}} = \left[\frac{x + \sqrt{x^2 - 1}}{\sqrt{x^2 - 1}} \right] \frac{1}{x + \sqrt{x^2 - 1}} = \frac{1}{\sqrt{x^2 - 1}}$$

Therefore

$$\boxed{\frac{d}{dx}[\cosh^{-1} x] = \frac{1}{\sqrt{x^2 - 1}}}$$

2.3 Inverse Hyperbolic Tangent Function

$y = \tanh^{-1} x \iff x = \tanh y$. Clearly $R = \mathbb{R}$. To determine the domain of definition, let $u = e^y$.

$$x = \frac{u - \frac{1}{u}}{u + \frac{1}{u}} = \frac{u^2 - 1}{u^2 + 1}$$

Therefore $xu^2 + x = u^2 - 1 \Rightarrow u^2(x - 1) = -1 - x \Rightarrow u^2 = \frac{1+x}{1-x} \Rightarrow e^{2y} = \frac{1+x}{1-x}$. Now we can write y as:

$$2y = \ln \left(\frac{1+x}{1-x} \right) \Rightarrow y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

The necessary condition for the domain to be $(-1, 1)$ is $|x| < 1$.

2.3.1 Derivative of Hyp. Tan

$$y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) = \frac{1}{2} [\ln(1+x) - \ln(1-x)] \Rightarrow \frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{1+x} + \frac{1}{1-x} \right] = \frac{1}{1-x^2}$$

Therefore

$$\boxed{\frac{d}{dx} [\tanh^{-1} x] = \frac{1}{1-x^2}}$$

2.4 Inverse Hyperbolic Cotangent Function

$y = \coth^{-1} x \iff x = \coth y$. Clearly $R = \mathbb{R}$. To determine the domain of definition

$$x = \frac{e^y + e^{-y}}{e^y - e^{-y}} = \frac{u + \frac{1}{u}}{u - \frac{1}{u}} \Rightarrow x = \frac{u^2 + 1}{u^2 - 1} \Rightarrow xu^2 - x = u^2 + 1 \Rightarrow u^2 = \frac{1+x}{x-1}$$

This implies that

$$e^{2y} = \frac{1+x}{x-1} \Rightarrow y = \frac{1}{2} \ln \left(\frac{1+x}{x-1} \right)$$

Now $D = (-\infty, -1) \cup (1, \infty)$

2.4.1 Derivative of Hyp. Cot

$$y = \frac{1}{2} \ln \left(\frac{1+x}{x-1} \right) = \frac{1}{2} [\ln(1+x) - \ln(x-1)] \Rightarrow \frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{1+x} - \frac{1}{x-1} \right] = \frac{-1}{x^2-1}$$

In general

No.	The Hyp. Inverse Function	Derivative
(1)	$\frac{d}{dx}[\sinh^{-1}(u(x))]$	$\frac{u'(x)}{\sqrt{u^2 + 1}}$
(2)	$\frac{d}{dx}[\cosh^{-1}(u(x))]$	$\frac{u'(x)}{\sqrt{u^2 - 1}}$
(3)	$\frac{d}{dx}[\tanh^{-1}(u(x))]$	$\frac{u'(x)}{1 - u^2}$
(4)	$\frac{d}{dx}[\coth^{-1}(u(x))]$	$\frac{u'(x)}{1 - u^2}$
(5)	$\frac{d}{dx}[\operatorname{sech}^{-1}(u(x))]$	$\frac{-u'(x)}{ u(x) \sqrt{1 - u^2}}$
(6)	$\frac{d}{dx}[\operatorname{csch}^{-1}(u(x))]$	$\frac{-u'(x)}{ u(x) \sqrt{u^2 + 1}}$